

Note 3: The PIVP Compiler Problem

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1 The Problem

Computing $\zeta(3) \in R_{\text{RTCRN}}$ requires a polynomial PIVP with rational initial conditions whose output converges to $\zeta(3)$. The known ODE system for $\zeta(3)$ (via the Beukers–Apéry integral $\int_0^\infty t^2/(e^t - 1) dt$) requires transcendental initial conditions involving e and $1/(e - 1)$.

Definition 1 (PIVP Compiler). A **PIVP compiler** is a systematic method that transforms a sequential composition of PIVPs:

1. Block A: a PIVP with rational ICs computing $\alpha = \lim_{t \rightarrow \infty} u(t)$ (e.g., $\alpha = e$),
2. Block B: a PIVP with IC depending on α computing $\beta = \lim_{t \rightarrow \infty} z(t)$ (e.g., $\beta = \zeta(3)$),

into a **single** polynomial PIVP with rational ICs whose output converges to β .

2 Approaches Explored

2.1 Gate Chain (Time-Scale Separation)

Idea. Run Block A freely. Gate Block B’s evolution with a chain of logistic variables $g'_k = g_{k-1}^2(1 - g_k)$. Block B only starts after Block A has approximately converged.

Properties.

- Polynomial RHS, rational ICs, bounded trajectories. ✓
- Gate chain of depth k reduces transient error geometrically. ✓
- Error is nonzero for any *fixed* k — only approximate. ✗

Why it fails. Block B computes an integral $I = \int_0^1 f(x, u(\tau(x))) dx$, where $u(\tau(x))$ is Block A’s output at the time Block B visits x . Since $u \neq \alpha$ during early integration, $I \neq \int_0^1 f(x, \alpha) dx$. The error is

$$\varepsilon = \int_0^1 [u(\tau(s)) - \alpha] \cdot \frac{\partial f}{\partial u} \cdot \frac{1}{1+s} ds,$$

which is bounded but nonzero for any finite gate depth.

Observation 1 (Numerical). *For the test problem $e \cdot \ln 2$: gate depth 1 gives 13.5% error, depth 7 gives 0.01% error. Error $\approx C \cdot 3^{-k}$ for depth k .*

2.2 Direct Replacement

Idea. In the Apéry $\zeta(3)$ system (12 variables), replace the static constants $e, C = 1/(e-1), D = e/(e-1)$ with dynamic estimators $E_1(t) \rightarrow e, E_2(t) \rightarrow 1/(e-1), E_3(t) \rightarrow e/(e-1)$.

Result. Fails. The target variable $s_3(t)$ does not converge to $\zeta(3)$. “The time-varying nature of the E_k functions interacts with the main system in a non-trivial way” (replacement.tex, DNA32 project).

Why it fails. Same mechanism as the gate chain: the main computation “sees” the wrong parameter values during its transient phase, and the accumulated error persists.

2.3 Explosion as Infinity

Idea. Use a finite-time blow-up $v' = v^2, v(0) = 1$ (explosion at $t = 1$) to create “infinite effective time” for Block A. Block A converges to α by $t = 1^-$. Then Block B runs in $t \in (1, 2)$, using a second explosion at $t = 2$.

Properties.

- $v = 1/(1-t)$, effective time $s_1 = -\ln(1-t) \rightarrow \infty$ as $t \rightarrow 1^-$.
- In s_1 -time, u converges to e . ✓

Obstruction 1: Freezing. After DNA32 resolution ($w = 1/v = 1-t$), the variable $u(t) = e^t$ passes through e at $t = 1$ but does not freeze. Freezing requires a polynomial f with $f(e) = 0$; since e is transcendental, no such polynomial exists.

Obstruction 2: Stability Reversal. For the logistic test $u' = u(1-u)/w$ with $w = 1-t$: the fixed point $u = 1$ is stable for $w > 0$ (before explosion) but *unstable* for $w < 0$ (after). The sign change in w reverses the eigenvalue. Numerically: $u(2) \approx 10^6$.

Obstruction 3: Snapshot Problem. In a concurrent ODE, Block B reads $u(t)$ (current value), not $u(1)$ (snapshot). Even with the explosion giving infinite effective time at $t = 1$, Block B’s integration variable sweeps $0 \rightarrow 1$ *during* the explosion, and u changes during this sweep.

3 Two Types of Limits

Definition 2. A PIVP computation is of **fixed-point type** if $\lim_{t \rightarrow \infty} y(t)$ converges to a globally attracting fixed point of the vector field (e.g., $u' = u(1-s), s' = 1-s$ gives $u \rightarrow e$).

A PIVP computation is of **integral type** if $\lim_{t \rightarrow \infty} y(t) = \int_0^1 f(x) dx$, computed by accumulation along a trajectory (e.g., $I' = f(x)(1-x), x' = 1-x$).

Observation 2. *Fixed-point computations are robust to parameter errors: the attractor “washes out” transient errors. Integral computations accumulate errors from the entire trajectory. This distinction is critical for PIVP compilation.*

4 The Lyapunov Adaptive Control Approach

Proposed framework (gemini_draft.tex, DNA32 project):

1. **GPAC Hierarchy (GPAC-H).** Floor 0: rational ICs. Floor n : uses Floor $n-1$ constants.
2. **Certainty Equivalence.** Replace unknown constants with dynamic estimators.
3. **Lyapunov Design.** Find V (function of all state variables and estimation errors) with $V' \leq 0$.

4. **Dynamic Adaptation Gains.** Adaptation rates are polynomial state variables.

Key property. If $V' \leq 0$ can be achieved with polynomial dynamics and rational ICs, the compiled system converges to the correct value *regardless of transient estimator errors*. This would unify fixed-point and integral type computations.

Open question. Can a suitable Lyapunov function be constructed for the 12-variable Apéry system?

5 Summary

Approach	Poly RHS?	Rational ICs?	Zero error?
Gate chain (depth k)	Yes	Yes	No ($\sim 3^{-k}$)
Direct replacement	Yes	Yes	No (diverges)
Explosion + resolution	Partially	Yes	No (unfreezing)
Lyapunov adaptive	Yes (if exists)	Yes	Yes (if $V' \leq 0$)

The PIVP compiler problem reduces to: **construct a Lyapunov function V for the coupled estimator-plus-main system, with $V' \leq 0$ achieved by polynomial dynamics.** Alternatively, find a fundamentally different compilation scheme (e.g., via explosions) that achieves zero error.